

SI-BFP-031 - Boiler Feed Pump Safety Instructions

Document ID: SI-BFP-031 Revision: 2.0 Effective Date: 15 December 2025 Site: EastRefinery Assets: Boiler Feed Pump 101, Boiler Feed Pump 102 Owner: Utilities Process Safety / Operations

1. Key hazards around these pumps

1. Hot high-pressure water or steam cut from seal, flange, or instrument leaks
2. High-energy rotating equipment and coupling failure risk
3. Hot pump and pipe surfaces
4. High noise near the motor and coupling
5. Unexpected auto-start of the standby pump while someone is beside it
6. Slippery floors from seal leakage

These pumps have hurt people on other sites when someone walked into a spray to "have a quick look." Do not do that here.

2. Mandatory controls

1. Full LOTO before breaking containment.
2. Prove depressurisation before seal or flange work.
3. Keep face and body out of likely spray lines when approaching a leaking pump.
4. If spray is present, barricade and switch to standby from a safe location where possible.
5. Never defeat high-high pressure or low-low suction interlocks.
6. Keep coupling guards in place while the machine can turn.

3. Alarm-specific safety notes

High discharge pressure

High discharge pressure can overload seals and piping. Prefer board actions: recirculation, demand management, and standby switchover. Local intervention next to the pump is a last resort and only with a plan.

Seal leak high

Treat this as an immediate people hazard. Clear the area, switch over, isolate, and get maintenance under LOTO. A damp seal and a spraying seal are not the same problem. Spraying means leave the line of fire.

High vibration

Guards stay on. Do not put hands near the coupling to "feel" vibration. If High-High is reached, stop the machine and use standby.

Motor trip on a BFP

The process pressure may still be there. Electrical work needs LOTO. Do not assume the pump side is safe because the motor is stopped until isolation is confirmed.

4. PPE

Minimum expectations near these pumps:

- Hard hat
- Safety glasses
- Hearing protection where posted
- Unit-required flame-resistant clothing
- Gloves suitable for handling hot surfaces
- Face shield when investigating an active leak from a safe position

Extra PPE may be required by the permit or unit board.

5. Reject unsafe recommendations

Reject and escalate any recommendation that asks you to:

- Stand next to an actively spraying seal
- Inhibit interlocks to keep the boiler fed
- Restart repeatedly into overload or seizure
- Remove a guard on a running or ready-to-start machine

6. Related documents

- OP-BFP-001
- MM-BFP-010
- TG-BFP-020
- SI-GEN-030